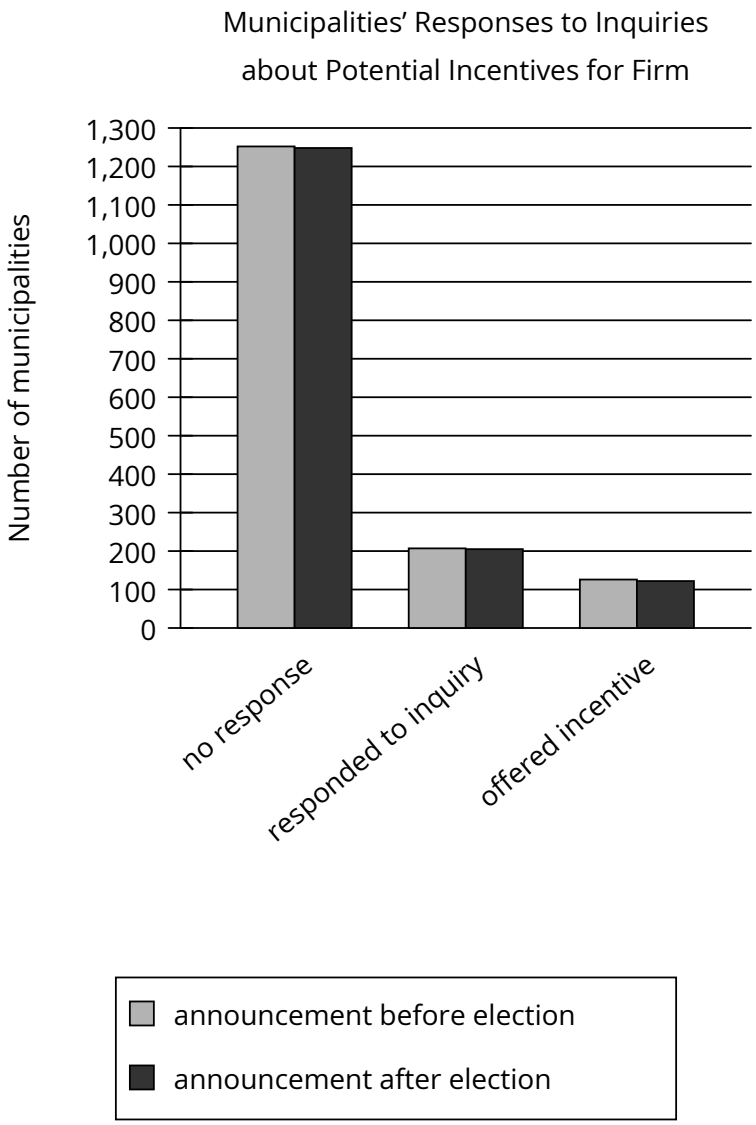
Question ID: a15b3219

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In the United States, firms often seek incentives from municipal governments to expand to those municipalities. A team of political scientists hypothesized that municipalities are much more likely to respond to firms and offer incentives if expansions can be announced in time to benefit local elected officials than if they can't. The team contacted officials in thousands of municipalities, inquiring about incentives for a firm looking to expand and indicating that the firm would announce its expansion on a date either just before or just after the next election.

Which choice best describes data from the graph that weaken the team's hypothesis?

Answer

A. A large majority of the municipalities that received an inquiry mentioning plans for an announcement before the next election didn't respond to the inquiry.

B. The proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ across the announcement timing conditions.

C. Only around half the municipalities that responded to inquiries mentioning plans for an announcement before the next election offered incentives.

D. Of the municipalities that received an inquiry mentioning plans for an announcement date after the next election, more than 1,200 didn't respond and only around 100 offered incentives.

Correct Answer: B

Rationale

Choice B is the best answer. The lighter bars show what happened when the announcement was to come before the election, and the darker bars show what happened when the announcement was to come after the election. For all three of the outcomes, the light and dark bars are virtually the same, demonstrating that the announcement timing didn't actually make a difference.

Choice A is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice C is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement after election" data for comparison. Choice D is incorrect. This accurately describes some data from the graph, but it doesn't weaken the hypothesis. It doesn't include the "announcement before election" data for comparison.

Question ID: 0d7f4966

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Jean-Bernard Caron and colleagues recently discovered a cache of jellyfish fossils in the Burgess Shale, a site in the Canadian Rockies that is rich in fossils from the Cambrian period (over 500 million years ago). Caron and colleagues claim that these are the oldest jellyfish fossils ever discovered. In the past twenty years, two sites in China and the United States have yielded fossils of a similar age that some experts believe are most likely jellyfish due to their shapes and the appearance of projecting tentacles. But Caron and colleagues argue that the apparent tentacles are in fact the comb rows of ctenophores, gelatinous animals that are only distantly related to jellyfish.

Which statement, if true, would most directly weaken the claim by Caron and colleagues about the fossils found in China and the United States?

Answer

A. Sites in the Canadian Rockies from later periods than the Cambrian period have yielded fossils that have been conclusively identified as ctenophore fossils.

B. The fossils found in China and the United States are so poorly preserved that though they cannot be conclusively identified as jellyfish, they cannot be conclusively identified as ctenophores either.

C. While ctenophore fossils have been discovered in China and the United States, they have never been discovered in the Burgess Shale.

D. The fossils discovered by Caron and colleagues in the Burgess Shale were better preserved than the fossils discovered by other researchers in China and the United States.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a statement that, if true, would most directly weaken Caron and colleagues' claim that the apparent tentacles in the Chinese and American fossils are actually ctenophore comb rows. If the fossils are so poorly preserved that they cannot be conclusively identified as either organism, neither the claim that they are jellyfish nor, as Caron claims, that they are ctenophores would be supported.

Choice A is incorrect. Caron's claim is that fossils from the US and China are ctenophores, not jellyfish. These fossils are said to be "of a similar age" to the Cambrian fossils found in the Canadian Rockies. And nothing in the text or this choice suggests that the presence or absence of ctenophores after the Cambrian would have any bearing on whether the Cambrian fossils from the US and China are ctenophores. Choice C is incorrect. Caron's claim is that fossils from the US and China are ctenophores, not jellyfish. Nothing in the text suggests that the presence or absence of ctenophores in the Burgess Shale (in Canada) would affect whether the fossils found in the US and China are ctenophores. Choice D is incorrect. Caron's claim is that fossils from the US and China are ctenophores, not jellyfish. Although fossil quality is a plausible issue for the research described in the text, nothing in the text or this choice suggests that the fossils from US and China would have been too poorly preserved for proper identification.

Question ID: 068f939b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The ancient Greek concept of “mimesis,” a term used in the works of Plato, Aristotle, and other Greek philosophers in discussions of representational art—visual, performance, or literary art that aims to depict the real world—is a foundational concept of the Western philosophy of aesthetics. Mimesis is typically translated as “imitation” in modern editions of ancient Greek texts, but scholar Stephen Halliwell warns that this is overly reductive: “imitation” implies that art merely copies—and is thus by definition entirely derivative of—a reality that exists outside and prior to the work of art, and translating “mimesis” thusly obscures the multifaceted ways in which the ancient Greeks understood the relationship between art and reality.

Which statement, if true, would most directly support the claim by Halliwell presented in the text?

Answer

A. One of the earliest appearances of mimesis’s root word, *mimos*, can be found in an ancient Greek tragedy in reference to dramatic impersonation, and the *mim-* root came to be generally associated with the musical and poetic arts by the fifth century BCE.

B. Both Plato’s and Aristotle’s theorizations of mimesis examine the psychological effects that works of art induce in the viewer or listener.

C. Although several of Plato’s earliest philosophical works discuss aesthetic ideas, the term “mimesis” doesn’t appear in Plato’s discussions of art until *Cratylus*, a relatively late work.

D. Although Plato’s writings typically characterize representational art as an inferior reflection of the physical world, Aristotle suggests that mimesis can refer to art’s capacity to envision hypothetical conditions that could, but don’t yet, exist.

Correct Answer: D

Rationale

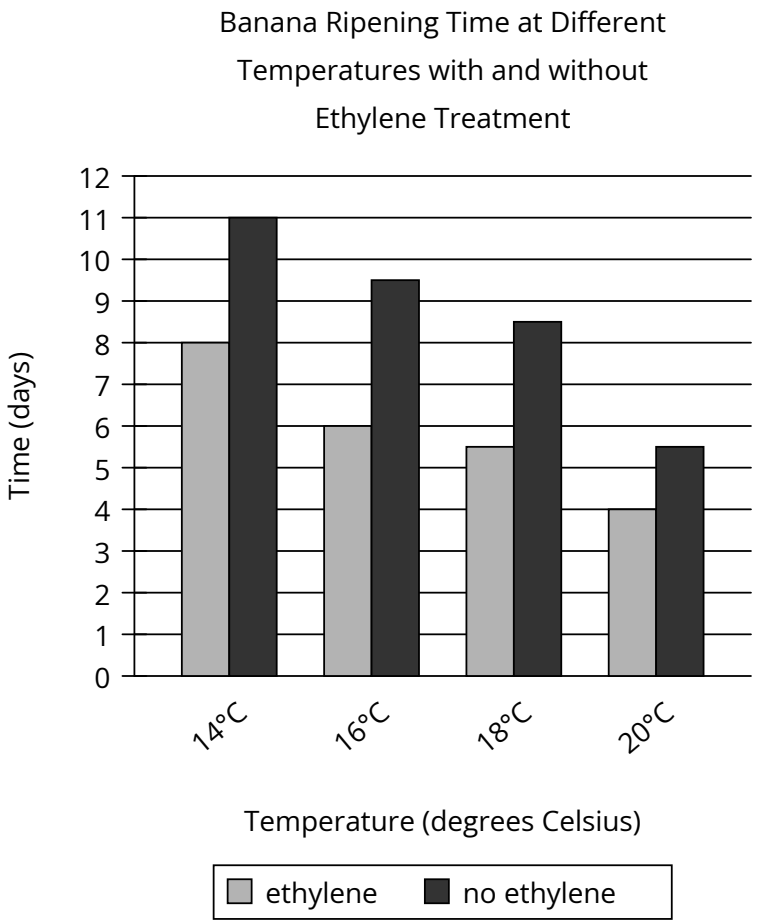
Choice D is the best answer because it presents a statement that would support Halliwell’s claim that "imitation" is an overly simplified translation of the word "mimesis" in the context of ancient Greek philosophical discussions of representational art and that, because it suggests a view of art as merely copying things that exist in reality, the translation obscures the fact that Plato, Aristotle, and other ancient Greeks understood the relationship between art and reality in varied ways. If Plato’s works tend to treat representational art as an inferior or inadequate reflection of the physical world (consistent with the idea of mere imitation), while Aristotle’s works suggest that the term "mimesis" can refer to art’s ability to show conditions that don’t currently exist but could exist (going beyond imitation into novel creation), that evidence would support the claim that the term "mimesis" means more than just "imitation" and that ancient Greek philosophers held different views of the relationship between art and reality.

Choice A is incorrect because information about the root word *mimos* first being used in drama and the root *mim-* coming to be associated with music and poetry wouldn’t indicate anything about the meaning of the specific term "mimesis" as it was used by ancient Greek philosophers in discussions of representational art; thus, the information would have no bearing on Halliwell’s claim that a common translation of the specific term is an oversimplification. Choice B is incorrect because the issue of art’s psychological effects on audiences gets at how people respond to works of art instead of how art itself is related to reality, so the idea that Plato and Aristotle both addressed such effects in their considerations of mimesis wouldn’t have any bearing on Halliwell’s claim that a common translation of that term oversimplifies the relationship between art and reality. Further, this idea would touch on one general similarity in approaches to mimesis instead of illustrating the multifaceted, or varied, ways Halliwell claims ancient Greek philosophers understood art’s relation to reality. Choice C is incorrect because the idea that Plato discussed aesthetics in his early works but didn’t use the term "mimesis" until later doesn’t give any indication of how Plato or any other ancient Greek philosophers understood the relation between art and reality, so it would have no bearing on Halliwell’s claim that a common translation of the term "mimesis" is oversimplified and fails to reflect the varied ways the philosophers understood that relationship.

Question ID: 040583a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



A student is conducting an experiment to test the effect of temperature and ethylene treatment on the ripening speed of bananas. The student treated some bananas with ethylene while leaving others untreated, then allowed both types of bananas to ripen at one of four different temperatures. Comparing the data for bananas with and without ethylene, the student concluded that _____

Which choice most effectively uses data from the graph to complete the student's conclusion?

Answer

A. 20°C is the ideal temperature at which to store bananas to slow ripening time.

B. for those bananas that were not treated with ethylene, differences in temperature were not associated with absolute differences in ripening time.

C. bananas treated with ethylene ripen faster at 14°C and 16°C than at 18°C and 20°C.

D. ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses data from the graph to complete the student's conclusion about banana ripening time with and without ethylene at different temperatures. The graph shows that at 20°C, the gap between the two bars showing ripening times for ethylene-treated bananas and untreated bananas crosses fewer than 2 gridlines (from about 4 days for ethylene-treated bananas to about 5.5 days for untreated bananas). Meanwhile, the graph shows that at 14°C, 16°C, and 18°C, the gap between the bars crosses more than 2 gridlines (from about 8 days for treated bananas to about 11 days for untreated bananas at 14°C; from about 6 days for treated bananas to about 9.5 days for untreated bananas at 16°C; and from about 5.5 days for treated bananas to about 8.5 days for untreated bananas at 18°C). Since the gap between the bars at each of these temperatures crosses more than 2 gridlines, and since each of these gaps is larger than the gap between the bars at 20°C, it can be concluded that ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

Choice A is incorrect because the graph shows that ethylene-treated bananas stored at 20°C ripen more quickly than ethylene-treated bananas stored at the other temperatures do (about 4 days at 20°C vs. about 5.5, 6, and 8 days at 18°C, 16°C, and 14°C, respectively) and that untreated bananas stored at 20°C ripen more quickly than untreated bananas stored at the other temperatures do (about 5.5 days at 20°C vs. about 8.5, 9.5, and 11 days at 18°C, 16°C, and 14°C, respectively). The information in the graph therefore indicates that storing bananas at 20°C speeds up ripening time relative to storing bananas at the other temperatures shown, not that this storage temperature slows ripening time. Choice B is incorrect because the graph shows that as temperature increases, the ripening time of untreated bananas decreases, from about 11 days at 14°C to about 5.5 days at 20°C, with no exceptions to this trend. The graph therefore shows that differences in temperature were associated with absolute differences in ripening time, not that there was no association between differences in temperature and differences in ripening time. Choice C is incorrect because the graph shows that ripening times of ethylene-treated bananas at 14°C and 16°C were about 8 and 6 days, respectively, which is greater than, not less than, ripening times of ethylene-treated bananas at 18°C and 20°C, which were about 5.5 and 4 days, respectively. In other words, bananas treated with ethylene ripen more slowly, not faster, at 14°C and 16°C than at 18°C and 20°C.

Question ID: 7a1877be

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Nucleobase Concentrations from Murchison Meteorite and Soil Samples in Parts per Billion

Nucleobase	Murchison meteorite sample 1	Murchison meteorite sample 2	Murchison soil sample
Isoguanine	0.5	0.04	not detected
Purine	0.2	0.02	not detected
Xanthine	39	3	1
Adenine	15	1	40
Hypoxanthine	24	1	2

Employing high-performance liquid chromatography—a process that uses pressurized water to separate material into its component molecules—astrochemist Yashiro Oba and colleagues analyzed two samples of the Murchison meteorite that landed in Australia as well as soil from the landing zone of the meteorite to determine the concentrations of various organic molecules. By comparing the relative concentrations of types of molecules known as nucleobases in the Murchison meteorite with those in the soil, the team concluded that there is evidence that the nucleobases in the Murchison meteorite formed in space and are not the result of contamination on Earth.

Which choice best describes data from the table that support the team's conclusion?

Answer

A. Isoguanine and purine were detected in both meteorite samples but not in the soil sample.

B. Adenine and xanthine were detected in both of the meteorite samples and in the soil sample.

C. Hypoxanthine and purine were detected in both the Murchison meteorite sample 2 and in the soil sample.

D. Isoguanine and hypoxanthine were detected in the Murchison meteorite sample 1 but not in sample 2.

Correct Answer: A

Rationale

Choice A is the best answer. The researchers concluded that the meteorite's nucleobases weren't the result of soil contamination. Presence of nucleobases in the meteorite and not in soil provides evidence that those nucleobases likely didn't come from the soil.

Choice B is incorrect. This choice doesn't justify the conclusion. The researchers concluded that the meteorite's nucleobases weren't the result of soil contamination. If the nucleobases are present in both the soil and meteorite, then it's possible that these nucleobases came from the soil. Choice C is incorrect. This choice misreads the table. Purine was not detected in the soil sample. Choice D is incorrect. This choice misreads the table. Both isoguanine and hypoxanthine were detected in both Murchison meteorite samples.

Question ID: 04cbeca3

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In 1534 CE, King Henry VIII of England split with the Catholic Church and declared himself head of the Church of England, in part because Pope Clement VII refused to annul his marriage to Catherine of Aragon. Two years later, Henry VIII introduced a policy titled the Dissolution of the Monasteries that by 1540 had resulted in the closure of all Catholic monasteries in England and the confiscation of their estates. Some historians assert that the enactment of the policy was primarily motivated by perceived financial opportunities.

Which quotation from a scholarly article best supports the assertion of the historians mentioned in the text?

Answer

A. "At the time of the Dissolution of the Monasteries, about 2 percent of the adult male population of England were monks; by 1690, the proportion of the adult male population who were monks was less than 1 percent."

B. "A contemporary description of the Dissolution of the Monasteries, Michael Sherbrook's *Falle of the Religious Howses*, recounts witness testimony that monks were allowed to keep the contents of their cells and that the monastery timber was purchased by local yeomen."

C. "In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was swiftly completed by the summer."

D. "The October 1536 revolt known as the Pilgrimage of Grace had several economic motives: high food prices due to a poor harvest the prior year; the Dissolution of the Monasteries, which closed reliable sources of food and shelter for many; and rents and taxes throughout Northern England that were not merely high but predatory."

Correct Answer: C

Rationale

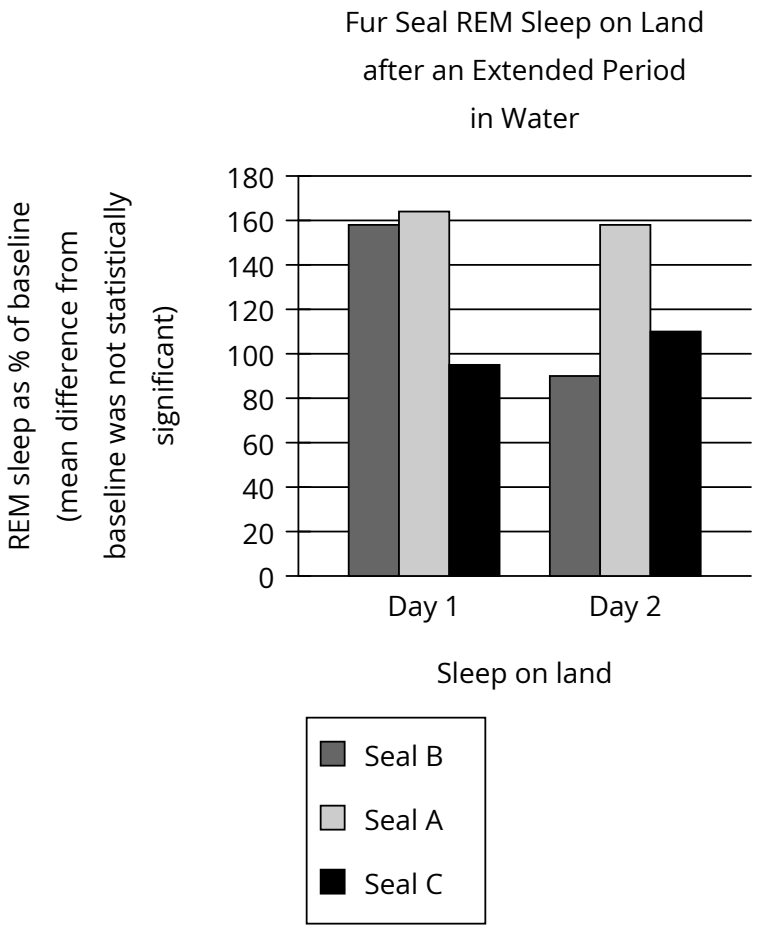
Choice C is the best answer. The fact that Henry VIII commissioned a survey of church holdings just before enacting the Dissolution of the Monasteries suggests that he was interested in the potential profits from claiming their assets. This supports the historians' assertion.

Choice A is incorrect. This choice describes a potential effect of the Dissolution of the Monasteries. The decrease in the proportion of monks in England isn't relevant to the question of Henry VIII's motivations. Choice B is incorrect. Details about how the monks were treated during the dissolution don't address Henry VIII's motivations for enacting the policy in the first place. That the monks could keep the content of their cells (their rooms) and sell off the timber they'd harvested has no impact on the value of the monasteries' estates—their land holdings. Choice D is incorrect. This choice mentions one impact that the Dissolution of the Monasteries contributed to, two years after it happened. But it doesn't help explain why Henry VIII might have wanted to enact the policy in the first place.

Question ID: ccb1ab92

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____

Which choice most effectively uses data from the graph to complete the text?

Answer

A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.

B. showed no significant differences from one another in baseline levels of REM sleep.

C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.

D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

Correct Answer: C

Rationale

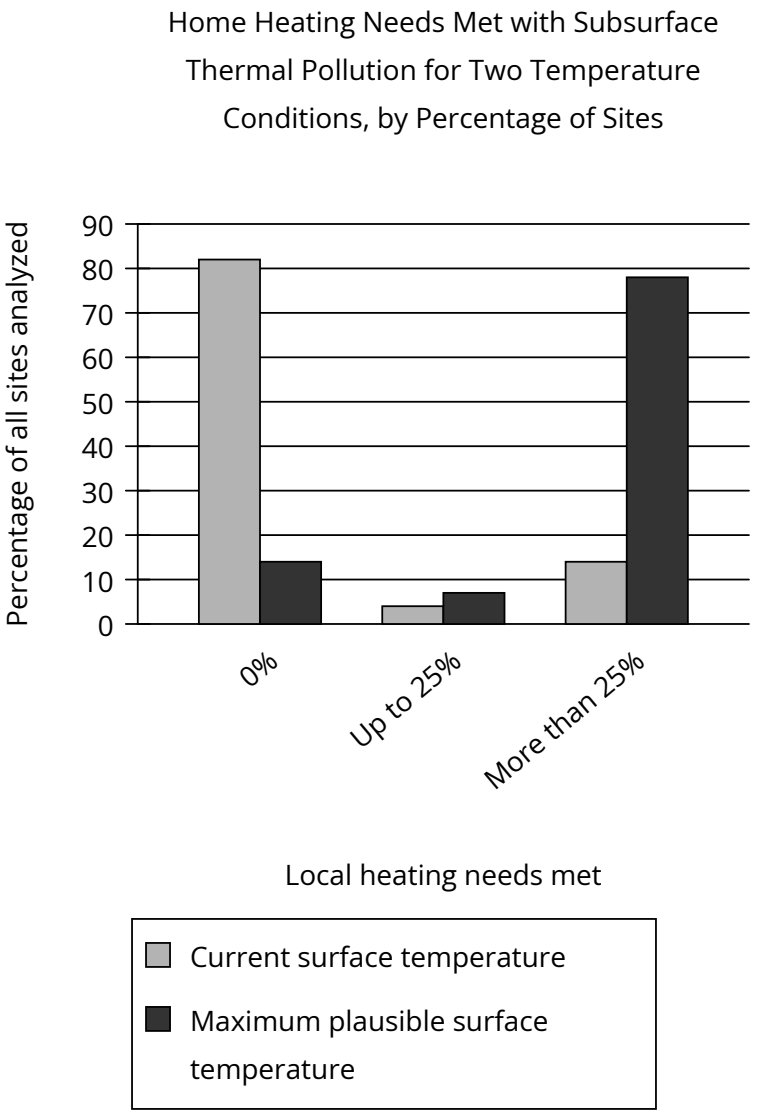
Choice C is the best answer. If REM sleep were homeostatically regulated in fur seals, then all the seals would compensate with REM levels significantly over baseline after going weeks without REM. We'd also expect the seals to maintain those elevated REM levels for some time. Since seals B and C return very quickly to baseline REM levels, this suggests that REM sleep in fur seals may not be regulated homeostatically.

Choice A is incorrect. This doesn't support the conclusion. If REM sleep were homeostatically regulated in fur seals, then we'd suspect the seals to sustain REM levels well above baseline for a prolonged period in order to compensate for weeks of REM deprivation while in the water. Whether or not there's a reduction in REM sleep from day 1 to day 2 doesn't tell us how REM sleep on those days relates to baseline, which is where our focus should be. Choice B is incorrect. The y-axis of this graph doesn't depict baseline levels of REM sleep, but rather shows REM sleep as a percent of baseline. Choice D is incorrect. The graph doesn't depict REM sleep while in the water for the seals in the study. Additionally, we're told fur seals get no REM sleep while in the water, which is significantly different to the values shown in the graph for after they return to land.

Question ID: be19faa1

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Urbanization, industrialization, and the warming climate create thermal pollution (excess heat) in the shallow subsurface soil. Susanne A. Benz and colleagues analyzed thousands of sites on three continents under one scenario in which surface temperature remains at the current level and under another in which the surface reaches the maximum plausible temperature. They then categorized each site according to the percentage of local home heating needs that could be met using this excess subsurface heat. The team concluded that if surface temperature approaches the maximum plausible level, the percentage of sites where thermal pollution could feasibly contribute to meeting home heating needs will increase.

Which choice best describes data in the graph that support Benz and colleagues' conclusion?

Answer

A. Under both temperature conditions, less than 10% of sites were in the up-to-25% group, but at the maximum plausible surface temperature, almost 80% of sites could have all their local heating needs met by thermal pollution.

B. At current surface temperatures, more than 80% of the sites have no need for supplemental local home heating from subsurface thermal pollution, but at the maximum plausible surface temperature, more than 70% of sites exhibit significantly greater home heating needs.

C. At current surface temperatures, more than 80% of sites can meet, at most, 25% of local home heating needs with subsurface thermal pollution, but at the maximum plausible surface temperature, more than 80% of sites can meet greater than 25% of local home heating needs.

D. At current surface temperatures, more than 80% of the sites cannot use subsurface thermal pollution to meet any portion of local home heating needs, but at the maximum plausible surface temperature, that percentage drops below 20%.

Correct Answer: D

Rationale

Choice D is the best answer. The researchers concluded that as we approach maximum plausible surface temperatures, there will be a larger percentage of sites where thermal pollution could contribute to meeting home heating needs. By showing that only a small percentage of homes can currently use thermal pollution for home heating, and that this percentage would grow much larger at maximum plausible surface temperatures, this choice supports the researchers' conclusion.

Choice A is incorrect. We do not know how many sites could have all (i.e., 100%) of their local heating needs met by thermal pollution, as the graph only classifies sites by whether "0%," "Up to 25%," and "More than 25%" of heating needs could be met. Choice B is incorrect. The graph is not depicting need for supplemental heating from thermal pollution, but rather potential to use thermal pollution for supplemental heating. Choice C is incorrect. The graph indicates that, at current surface temperatures, less than 10% of sites can meet 25% of local home heating needs and that more than 80% of sites cannot meet any local home heating needs.

Question ID: c83e0b43

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather depicts Alexandra Bergson as a person who takes comfort in understanding the world around her: _____

Which quotation from *O Pioneers!* most effectively illustrates the claim?

Answer

A. “She looked fixedly up the bleak street as if she were gathering her strength to face something, as if she were trying with all her might to grasp a situation which, no matter how painful, must be met and dealt with somehow.“

B. “She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring.“

C. “Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country.”

D. “Alexandra drew her shawl closer about her and stood leaning against the frame of the mill, looking at the stars which glittered so keenly through the frosty autumn air. She always loved to watch them, to think of their vastness and distance, and of their ordered march. It fortified her to reflect upon the great operations of nature, and when she thought of the law that lay behind them, she felt a sense of personal security.”

Correct Answer: D

Rationale

Choice D is the best answer because it most effectively uses a quotation from *O Pioneers!* to illustrate the claim that Alexandra Bergson takes comfort in understanding the world around her. In the quotation, Alexandra is described as enjoying looking at the stars and feeling a “sense of personal security” when she contemplates nature’s order and its governing laws. This suggests that Alexandra takes comfort in understanding the world around her.

Choice A is incorrect because the quotation expresses how Alexandra Bergson attempts to meet difficult situations with determination, not how she takes comfort in understanding the world around her. Choice B is incorrect because the quotation expresses “how much the country meant to” Alexandra Bergson, not how she takes comfort in understanding the world around her. In detailing some of the wildlife surrounding Alexandra, the quotation conveys that nature is important to her but not necessarily that it gives her comfort. Choice C is incorrect because the quotation describes Alexandra driving her wagon down a highway at night; it doesn’t describe how she takes comfort in understanding the world around her or address how she’s feeling as she drives off.

Question ID: dd1757fd

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

Answer

A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.

B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.

C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.

D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

Correct Answer: A

Rationale

Choice A is the best answer. While many networks can perform navigation tasks, or even mimic grid cells, it doesn't mean they're actually behaving like biological brains—this finding suggests that the rules that govern neural network behavior are completely unlike the way real brains work.

Choice B is incorrect. Although it mentions the rules that are programmed into the networks, this finding wouldn't clarify whether or not these rules have anything to do with the function of biological brains. Choice C is incorrect. This choice suggests that neural networks are modeled after multiple types of brain cells, which sidesteps the question of whether these rule-based networks are genuinely similar to biological brains. Choice D is incorrect. This choice doesn't address the key point of the claim, which is that the apparent similarity between neural networks and biological brains is only due to the rules programmed into the networks. It focuses on training tasks, not the originally programmed rules.

Question ID: 09f9edb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can't be easily categorized according to British or North American popular music genres, typically referring to such music as "world music." While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they'd previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng's claim?

Answer

A. Mbaqanga and quan họ developed independently of each other and have little in common musically.

B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.

C. Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or Vietnam.

D. Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each other.

Correct Answer: A

Rationale

Choice A is the best answer. Zheng's claim is that the idea of world music "homogenizes" (meaning makes similar) distinct kinds of music by reducing them to one category. In other words, Zheng thinks the concept of world music is a harmful oversimplification of diverse musical forms. To support this claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. If it's true that mbaqanga and quan họ developed separately and have little in common musically, then it wouldn't make sense to lump them into the same category.

Choice B is incorrect. If true, this wouldn't affect the claim. To support the claim, we need evidence that these musical traditions are so different from one another that they should not fall into the same category. A difference in popularity doesn't necessarily mean that the two musical traditions shouldn't be categorized together: instead, we need to know if the music itself is similar or different. Choice C is incorrect. If true, this wouldn't affect the claim. To support the claim, we need evidence that these musical traditions are so different from each other that they should not fall into the same category. This choice doesn't do that. Choice D is incorrect. If true, this would actually weaken the claim. Zheng thinks it's reductive or oversimplifying to put distinct musical traditions into a single category. But if mbaqanga and quan họ are similar to each other, then it would make sense to put them in the same category.

Question ID: 39e440e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues' conclusion?

Answer

A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.

B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.

C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.

D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

Correct Answer: D

Rationale

Choice D is the best answer. This finding, if true, would support the archaeologists' conclusion. Dense central settlements linked to smaller ones would provide evidence of cities and suburbs—in other words, "a form of urbanism."

Choice A is incorrect. This finding, if true, would weaken the archaeologists' conclusion. Widely separated, small settlements with jungle in between would support the long-held belief that the Casarabe culture "was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness." Choice B is incorrect. This choice wouldn't support the researchers' conclusion. These large cities are located outside Casarabe territory, which doesn't show evidence of Casarabe urbanism. Choice C is incorrect. This finding wouldn't support the archaeologists' conclusion. Scattered small farms in jungle clearings are not good evidence to support the existence of cities ("a form of urbanism").

Question ID: 156ff681

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

Answer

A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.

B. Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the transfer.

C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.

D. Some recipients of physical transfers received small amounts of money relatively frequently, while others received large amounts relatively infrequently.

Correct Answer: B

Rationale

Choice B is the best answer. This would weaken the explanation. If the recipients of electronic money immediately withdrew it all as physical money, then both kinds of recipients ended up spending physical money on food. So there must be some other explanation why those who initially received electronic money ate different kinds of food.

Choice A is incorrect. This wouldn't weaken the explanation. If anything, it actually supports it: it demonstrates that recipients of electronic money and recipients of physical money have different spending habits. Choice C is incorrect. This wouldn't weaken the explanation. The explanation we're testing this choice against is about the way that people might "conceive of and allocate" physical and electronic funds differently. This choice only focuses on the timing, not what they spend the money on. Choice D is incorrect. This would have no impact on the explanation. It doesn't tell us anything about possible differences between the spending habits of those who spend physical money and those who spend money electronically.

Question ID: e946a32e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Boldly mixing elements of poetry, fiction, drama, philosophy, and manifesto, Puerto Rican writer Giannina Braschi creates cross-genre literature that explores themes such as immigration and independence. Her works have inspired responses from individuals across different fields and in a wide range of formats, from musical compositions and a comic book to architecture and furniture design. In an essay, a student asserts that the production of these diverse creations by others is reflective of Braschi's own approach to crafting literature.

Which quotation from a scholarly review of Braschi's work best supports the student's claim?

Answer

A. "Braschi is the focus of a 2020 collection of essays in which fifteen scholars from seven different countries delved into the linguistic and structural patterns of her writings."

B. "Braschi's eagerness to push boundaries and blend genres within literature invites us to consider how other art forms might also engage with literature."

C. "Before settling in New York City, where she would go on to become a college professor, Braschi studied both literature and philosophy in several cities around the world."

D. "In addition to her creative literary works, Braschi has produced academic pieces analyzing writings by Miguel de Cervantes, Federico García Lorca, and other authors."

Correct Answer: B

Rationale

Choice B is the best answer because it presents the quotation that best supports the student's claim about Braschi. By describing how Braschi's blending of genres invites her audience to think about how other art forms could also engage with literature, the quotation supports the idea that the diversity of responses to Braschi's work reflects Braschi's own approach to creating literature.

Choice A is incorrect because the quotation describes scholars from different countries writing essays about Braschi's use of language in her writings; it doesn't address how Braschi's creation of cross-genre literature inspires diverse types of responses, which is the claim the student makes. Choice C is incorrect because the quotation focuses on the fact that Braschi studied in several different cities, which doesn't address the student's claim that Braschi's creation of cross-genre literature inspires diverse types of responses. Choice D is incorrect because the quotation lists some of the authors who Braschi has written academic works about, which is irrelevant to the student's claim that Braschi's creation of cross-genre literature inspires diverse types of responses.

Question ID: dc87adf4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Barchester Towers is an 1857 novel by Anthony Trollope. In the novel, Trollope's portrayal of Dr. Proudie underscores the character's exaggerated sense of his own abilities: _____

Which quotation from *Barchester Towers* most effectively illustrates the claim?

Answer

A. "It must not…be taken as proved that Dr. Proudie was a man of great mental powers, or even of much capacity for business, for such qualities had not been required in him."

B. "[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts, natural and acquired, as must be sure to recommend him to a yet higher notice."

C. "[Dr. Proudie's] residence in the metropolis, rendered necessary by duties thus entrusted to him, his high connexions, and the peculiar talents and nature of the man, recommended him to persons in power."

D. "[Dr. Proudie] was certainly possessed of sufficient tact to answer the purpose for which he was required without making himself troublesome."

Correct Answer: B

Rationale

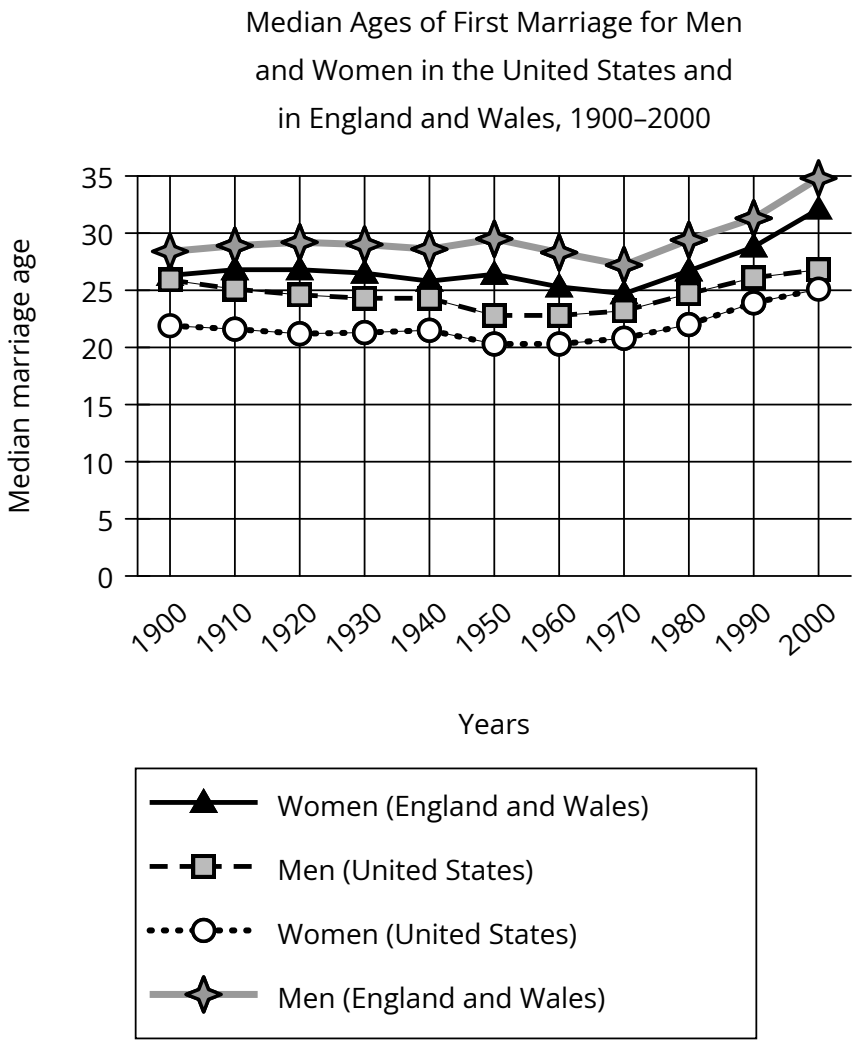
Choice B is the best answer. In this quotation, Dr. Proudie is described as "fondly flatter[ing] himself" that he has gifts that "must be sure to recommend him to a yet higher notice." In other words, he expects his skills to push him to greater fame and success. This implies an exaggerated sense of his own abilities, which matches the claim we're trying to support.

Choice A is incorrect. This quotation doesn't describe Proudie's view of himself, nor does it paint him in an especially flattering light. Instead, by saying his mental powers and business skill are not "proved," it implies that he is actually dim-witted and bad at business. Choice C is incorrect. This choice describes Proudie's closeness to power and importance, but it doesn't show what Proudie thinks of himself. Proudie is not describing himself or his abilities here. The narrator is. Choice D is incorrect. While this quotation offers Proudie very mild praise, it doesn't show what Proudie thinks of himself or his own abilities, which is what the claim focuses on.

Question ID: 53c6c179

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student's assertion?

Answer

A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.

B. In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.

C. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.

D. Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

Correct Answer: A

Rationale

Choice A is the best answer. The student concluded that country-specific factors influence changes to median age of first marriage. This choice describes a time period when the rate of change differed between countries, suggesting that country-specific factors may have played a role in these changes.

Choice B is incorrect. This choice misreads the graph. The median age of first marriage was consistently higher for men than for women in the United States during the time period depicted. Choice C is incorrect. The student concluded that country-specific factors influence changes to median age of first marriage. However, this choice doesn't provide any contrasts between countries and thus doesn't support the idea of country-specific factors influencing median age of first marriage. Choice D is incorrect. The students' conclusion is about changes that occurred during the 20th century. This choice provides broad information about the century as a whole, so it doesn't give insight into how median age at first marriage changed over time.

Question ID: df91532e

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In the “language nest” model of education, Indigenous children learn the language of their people by using it as the medium of instruction and socialization at pre-K or elementary levels. In their 2016 study of a school in an Anishinaabe community in Ontario, Canada, scholars Lindsay Morcom and Stephanie Roy (who are Anishinaabe themselves) found that the model not only imparted fluency in the Anishinaabe language but also enhanced students’ pride in Anishinaabe culture overall. Given these positive effects, Morcom and Roy predict that the model increases the probability that as adults, former students of the school will transmit the language to younger generations in their community.

Which finding, if true, would most strongly support the researchers’ prediction?

Answer

A. Anishinaabe adults who didn’t attend the school feel roughly the same degree of cultural pride as the former students of the school feel.

B. After transferring to the school, new students experience an increase in both fluency and academic performance overall.

C. As adults, former students of the school are just as likely to continue living in their community as individuals who didn’t attend the school.

D. As they complete secondary and higher education, former students of the school experience no loss of fluency or cultural pride.

Correct Answer: D

Rationale

Choice D is the best answer because it presents a finding that, if true, would support the researchers’ prediction about the language nest model of education. The text states that Morcom and Roy studied the effects of the language nest model of education on students at an Anishinaabe school, and they found that the model—which is used with students during pre-K or elementary school—increased students’ fluency in the Anishinaabe language and pride in Anishinaabe culture. The researchers predicted that the students’ positive early experiences with the Anishinaabe language would lead them to be more likely to later share the language with younger generations. If former students maintain full fluency and cultural pride after finishing secondary and higher education, it follows that they would be both able and motivated to share what they know with others; this would likely result in a higher probability of transmitting the language to younger generations, as the researchers predict.

Choice A is incorrect because finding that Anishinaabe adults who didn’t attend the school feel approximately the same degree of cultural pride as those adults who did attend wouldn’t support the researchers’ prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people. Choice B is incorrect because finding that new students experience increased performance in language fluency and academics would suggest that the school has a positive effect on students when they attended but wouldn’t reveal anything about those students’ later actions as adults (such as their likelihood of sharing their knowledge with younger generations). Choice C is incorrect because finding that Anishinaabe adults who attended the school are equally likely to stay in the community as adults who didn’t attend the school wouldn’t support the researchers’ prediction that former students will be more likely to share their knowledge with younger generations. This finding would identify a similarity between the groups rather than a factor that might make former students more likely than other adults to transmit the language to younger people.

Question ID: 44da37eb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

The variety of species with adaptations to produce toxins is matched by the variety of uses of those toxins: northern stargazers, for example, use toxins for defense, whereas tiger snakes use toxins for predation and skeleton shrimp use toxins for intraspecific competition. In fact, a species may have adaptations enabling it to produce a toxin with multiple uses. Finding that the venom used by the Panamanian scorpion *Centruroides granosus* to subdue prey also inhibits growth of the pathogenic bacteria *Escherichia coli*, Dumas Gálvez and colleagues conclude that the particular form of venom production observed in *C. granosus* may have propagated through the species because it mitigates risk during feeding in addition to enhancing predation success.

Which finding, if true, would most directly support Gálvez and colleagues' conclusion?

Answer

A. *E. coli* does not appear to be virulent for *C. granosus* even when transmitted from prey captured without the use of venom.

B. *E. coli* is frequently found in species preyed on by *C. granosus* and can survive exposure to the digestive juices of *C. granosus*.

C. *C. granosus* appears to be chemically sensitive to prey infected with *E. coli* and tends to favor uninfected individuals.

D. Exposure to *C. granosus* venom also inhibits the growth of nonpathogenic bacteria species common in the native environment of *C. granosus*.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that, if true, would most directly support Dumas Gálvez and colleagues' conclusion that the venom produced by the scorpion species *Centruroides granosus* helps mitigate, or reduce, risk during feeding in addition to enhancing predation success. According to the text, Gálvez and colleagues found that the venom used by *C. granosus* both subdues prey and inhibits the growth of *Escherichia coli*, a pathogen that, if ingested by *C. granosus*, would presumably cause disease unless neutralized in some way. If it were true that *E. coli* is commonly found in species preyed on by *C. granosus* and, moreover, can withstand *C. granosus*'s digestive juices after ingestion, this would suggest that *C. granosus* likely relies on another mechanism to neutralize *E. coli* to make *E. coli*-infected prey safe for consumption by the scorpion species. Given that, as the text states, *C. granosus*'s venom was found to inhibit the pathogen's growth, it therefore follows that the venom provides protection for *C. granosus* against *E. coli* that its digestive system wouldn't otherwise provide, making it reasonable to conclude that the trait may have spread through the species because it mitigates risk during feeding.

Choice A is incorrect because a finding that *E. coli* doesn't appear to be virulent, or dangerous, for *C. granosus* even when venom isn't used would weaken rather than strengthen Gálvez and colleagues' conclusion that a particular form of venom production spread in *C. granosus* in part because it helps reduce risk during feeing, since this finding would suggest that *E. coli* isn't actually a risk to *C. granosus* when consuming prey. Choice C is incorrect because a finding that *C. granosus* can detect and avoid consuming *E. coli*-infected prey in the first place would suggest that an ability other than venom production is the primary factor that reduces *C. granosus*'s risk when feeding, which would suggest there has been less evolutionary pressure to develop venom that provides protection; thus, this finding wouldn't clearly support the conclusion that *C. granosus*'s form of venom production spread in the species in part because it helps reduce risk during feeding. Choice D is incorrect because Gálvez and colleagues' conclusion focuses on *C. granosus* venom in relation to the risk of *E. coli*, and a finding that the venom also inhibits nonpathogenic (not disease-causing) bacteria species, which presumably don't pose a risk if consumed, wouldn't be relevant to the conclusion that *C. granosus*'s particular form of venom production spread in the species in part because it helps reduce risk during feeding.

Question ID: 626a1308

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Data Sources for Neptune Temperature Analysis

Instrument	Observatory	Data type	Observation years
TEXES (Texas Echelon Cross Echelle Spectrograph)	Gemini Observatory	spectroscopy	2007, 2019
T-ReCS (Thermal-Region Camera Spectrograph)	Gemini Observatory	infrared imaging	2007, 2010
LWS (Long Wavelength Spectrometer)	Keck Observatory	infrared imaging	2003
VISIR (VLT Imager and Spectrometer for mid-InfraRed)	European Southern Observatory	spectroscopy	2006

Julianne I. Moses and colleagues have reported that Neptune may have cooled significantly between 2003 and 2020. The team reached this conclusion by analyzing existing infrared imaging and spectroscopy data about the planet obtained from various instruments in different years. Of the team's sources listed in the table, the earliest example of spectroscopy data included in the analysis was obtained in _____

Which choice most effectively uses data from the table to complete the text?

Answer

A. 2007 using TEXES at the Gemini Observatory.

B. 2007 using T-ReCS at the Gemini Observatory.

C. 2006 using VISIR at the European Southern Observatory.

D. 2003 using LWS at the W.M. Keck Observatory.

Correct Answer: C

Rationale

Choice C is the best answer because it effectively completes the text's discussion of Moses and colleagues' analysis of changes in Neptune's temperature by using data from the table to identify the source of the earliest example of spectroscopy data included in this analysis. The text indicates that Moses and colleagues analyzed infrared imaging and spectroscopy data about Neptune obtained in different years by various instruments. The table lists the observatories where these instruments are housed, the type of data each instrument collected (infrared imaging or spectroscopy), and the years in which the instruments' observations were made. According to the table, only two of these instruments obtained spectroscopy data: VISIR at the European Southern Observatory, which made its observations in 2006, and TEXES at the Gemini Observatory, which made its observations in later years (2007 and 2019). Although two other instruments are also included in the table, they obtained infrared imaging data, not spectroscopy data. Therefore, the earliest example of spectroscopy data included in the researchers' analysis is the data obtained by VISIR at the European Southern Observatory in 2006.

Choice A is incorrect. Although the table indicates that TEXES at the Gemini Observatory collected spectroscopy data in 2007 and 2019, the table also indicates that this type of data was collected in 2006 by VISIR at the European Southern Observatory. Because the VISIR data were collected in the previous year, the data obtained in 2007 by TEXES are not the earliest example of spectroscopy data included in Moses and team's analysis of Neptune's temperature. Choice B is incorrect because the table indicates that T-ReCS at the Gemini Observatory collected infrared imaging data, not spectroscopy data. T-ReCS therefore cannot be the source of the earliest example of spectroscopy data that is included in Moses and colleagues' analysis of Neptune's temperature. Additionally, the T-ReCS data were collected in 2007, meaning that these data were obtained later than any type of data collected in 2006. Choice D is incorrect. Although the table indicates that data from LWS at the W.M. Keck

Observatory were obtained in 2003, making these data the earliest example of any type of data included in the analysis, the table also indicates that LWS collected infrared imaging data, not spectroscopy data. LWS therefore cannot be the source of the earliest example of spectroscopy data included in Moses and team's analysis of Neptune's temperature.

Question ID: a44bbd6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger's conclusion?

Answer

A. The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the sampling sites.

B. The streams contain several types of sediment that are not typically found in streams where the sorting effect has been demonstrated.

C. The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they take.

D. The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams downstream of the sampling sites.

Correct Answer: A

Rationale

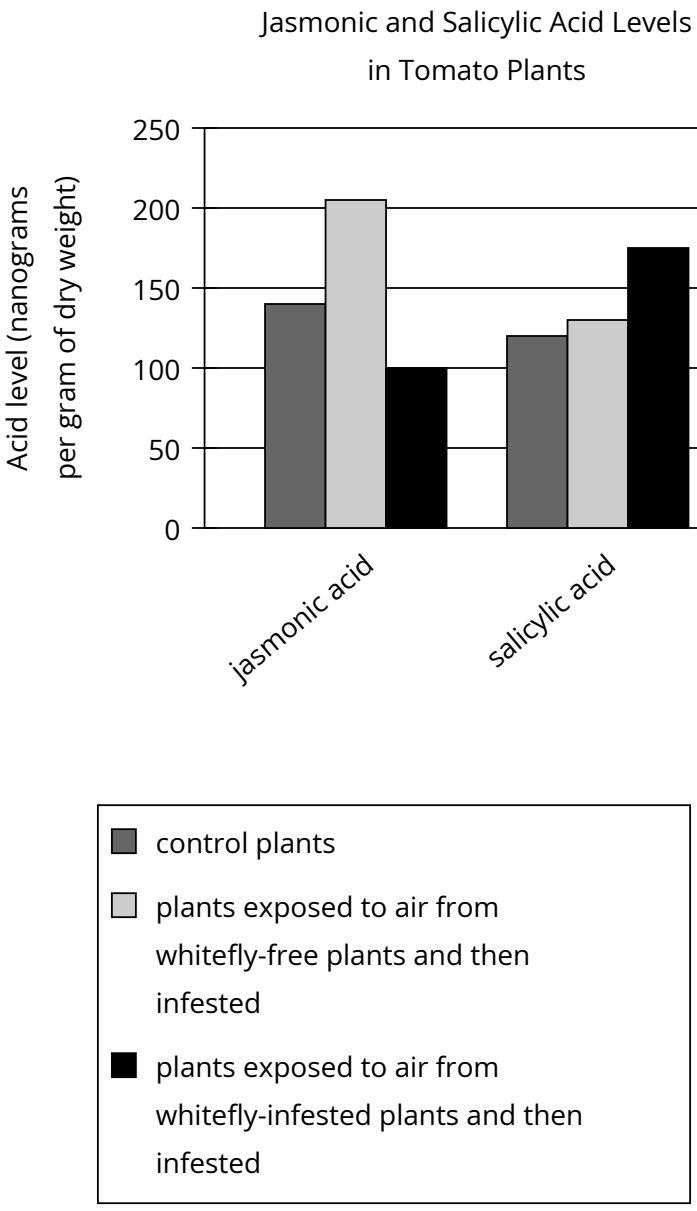
Choice A is the best answer. This finding would support the conclusion. If stream banks are collapsing into the water at multiple points, then sediment is getting into the water at those various points. This supports the conclusion that the inflow of sediment is very spread out.

Choice B is incorrect. This finding wouldn't support the conclusion. The conclusion is about the influx of sediment being "spatially diffuse," meaning spread out over a large area. The type of sediment wouldn't have an impact on the conclusions. Choice C is incorrect. This finding wouldn't support the conclusion. It doesn't say anything about the influx of sediment being "spatially diffuse" (spread out). Choice D is incorrect. This finding wouldn't support the conclusion. Any sediment that enters downstream of the sampling sites wouldn't end up in the samples, so it wouldn't affect the findings or the conclusion.

Question ID: d5da74be

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



In tomato plants, herbivory induces defensive production of jasmonic acid, while microbial infection induces defensive production of salicylic acid; plants also emit airborne chemicals to initiate the appropriate defense in nearby tomato plants. Researchers investigated the poor resistance tomato plants show to whitefly herbivory by exposing some plants to airborne chemicals from whitefly-free plants and others to airborne chemicals from whitefly-infested plants, then infesting both groups of plants with whiteflies. The researchers concluded that whiteflies induce tomato plants to emit chemicals that cause other tomato plants to preferentially defend against microbial infection even when under herbivorous attack.

Which choice best describes data from the graph that support the researchers' conclusion?

Answer

A. When plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid and more salicylic acid than did control

plants.

B. When plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid than salicylic acid, whereas when plants exposed to air from whitefly-free plants were infested, they produced about the same amount of jasmonic acid and salicylic acid.

C. When plants exposed to air from whitefly-free plants were infested, they produced both jasmonic acid and salicylic acid, whereas when plants exposed to air from whitefly-infested plants were infested, they exclusively produced salicylic acid.

D. When plants exposed to air from whitefly-infested plants were infested, they produced less jasmonic acid than did control plants, whereas when plants exposed to air from whitefly-free plants were infested, they produced more jasmonic acid and salicylic acid than did control plants.

Correct Answer: A

Rationale

Choice A is the best answer because it best describes data from the graph that support the researchers' conclusion about whitefly-induced defenses in tomato plants. According to the text, tomato plants produce airborne chemicals that prompt nearby tomato plants to increase their production of appropriate defensive chemicals, such as jasmonic acid in response to herbivory and salicylic acid in response to microbial infection. The text goes on to indicate that researchers concluded that whiteflies induce tomato plants to emit chemicals that cause nearby plants to defend against microbial infection instead of herbivory from whiteflies. The graph shows that control tomato plants (plants that weren't exposed to any chemicals from nearby tomato plants and weren't subsequently infested with whiteflies) produced about 140 nanograms of jasmonic acid per gram of dry weight and about 120 nanograms of salicylic acid per gram of dry weight, which indicates the baseline levels of these defensive chemicals in tomato plants. The graph also shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when subsequently infested by whiteflies, which is more than the amount produced by the control plants. This suggests that when infested with whiteflies, tomato plants that weren't previously exposed to air from nearby whitefly-infested plants begin producing higher levels of jasmonic acid to defend against the infestation. Finally, the graph also shows that plants exposed to air from whitefly-infested plants and then subsequently infested by whiteflies produced approximately 100 nanograms of jasmonic acid per gram of dry weight (which is less than control plants did) and approximately 175 nanograms of salicylic acid per gram of dry weight (which is more than control plants did). This suggests that whiteflies cause tomato plants to emit chemical signals that decrease nearby plants' defenses against whitefly herbivory relative to baseline levels and redirect their defenses toward microbial infection, thereby supporting the researchers' conclusion that whiteflies cause tomato plants to emit airborne chemical signals that induce nearby plants to preferentially defend against microbes even when experiencing herbivory from whiteflies.

Choice B is incorrect. Although the graph shows that tomato plants exposed to whitefly-infested plants produced less jasmonic acid than salicylic acid (about 100 nanograms of jasmonic acid per gram of dry weight vs. about 175 nanograms of salicylic acid per gram of dry weight), and although this might suggest that those plants are therefore better defended against microbial attack than against herbivory, this wouldn't be sufficient to conclude that this is the case: without reference to the relative levels of defensive chemicals produced by the plants in the other experimental conditions, it would be impossible to conclude that exposure to air from whitefly-infested plants either induces or diminishes these plants' defenses to different kinds of threats. Furthermore, the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight and about 130 nanograms of salicylic acid per gram of dry weight, not that they produced the same amount of the two acids. Choice C is incorrect because it inaccurately describes the data in the graph. The graph shows that plants in all three conditions produced both jasmonic and salicylic acid, not that plants exposed to air from whitefly-infested plants produce no jasmonic acid at all. Choice D is incorrect. Although the graph shows that plants exposed to air from whitefly-infested plants produced about 100 nanograms of jasmonic acid per gram of dry weight when infested, which is less than the amount produced by control plants (about 140 nanograms per gram of dry weight), and although the graph shows that plants exposed to air from whitefly-free plants produced about 205 nanograms of jasmonic acid per gram of dry weight when infested (which is more than the amount produced by control plants) and about 130 nanograms of salicylic acid per gram of dry weight (which is more than the approximately 120 nanograms per gram of dry weight produced by control plants), this wouldn't support the researchers' conclusion that plants exposed to air from whitefly-infested plants preferentially defend against microbial infection—that is, that they increase their levels of salicylic acid and decrease their levels of jasmonic acid relative to control plants. Rather, the data would suggest that exposure to air from whitefly-infested plants reduces plants' defenses against herbivory, but these data wouldn't indicate that whiteflies induce plants to redirect their defenses to guard against microbial infection. Furthermore, the data concerning plants exposed to air from whitefly-free plants merely indicate that these plants increase their defenses against both herbivory and microbial infection when infested by whiteflies.

Question ID: 63e7799d

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In vertical inheritance, parents pass genes to their offspring, but in horizontal transfer (HT), one species, often bacteria, passes genetic material to an unrelated species. In a 2022 study, herpetologist Atsushi Kurabayashi and his team investigated HT in multicellular organisms—namely, snakes and frogs in Madagascar. The team detected *BovB*—a gene transmitted vertically in snakes—in many frog species. The apparent direction of gene transfer seems counterintuitive because frogs usually don't survive encounters with snakes and so wouldn't be able to transmit the newly acquired gene to offspring, but the team concluded that *BovB* is indeed transmitted from snakes to frogs, either directly or indirectly, via HT.

Which finding, if true, would most directly support the team's conclusion?

Answer

A. *BovB* can be transmitted across frog species through HT.

B. Parasites known to feed on species of snakes and frogs in which the *BovB* gene occurs also carry *BovB*.

C. *BovB* cannot be reliably transmitted from a snake species to bacteria that are usually encountered by frog species.

D. Frog species with *BovB* show few discernible advantages as compared with frog species that do not carry *BovB*.

Correct Answer: B

Rationale

Choice B is the best answer. If there are parasites that feed on both snakes and frogs, they could carry *BovB* from snakes to frogs. HT could occur "indirectly" through those encounters, which frogs are more likely to survive than snake encounters.

Choice A is incorrect. The team's conclusion specifically argues that *BovB* is transmitted from snakes to frogs via HT, and this choice doesn't mention snakes. Choice C is incorrect. The team argues that *BovB* is transmitted from snakes to frogs via HT, and this answer choice discusses a scenario in which the gene is not transmitted. Choice D is incorrect. Whether or not frog species with *BovB* are advantaged isn't relevant to the team's conclusion about how the gene is transmitted.

Question ID: 4a07be59

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Neurobiologists Laura Cuaya, Raúl Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs. The researchers monitored the brain activity of Joey (an Australian shepherd), Mini (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech.

Which finding from the study, if true, would most directly support the team's conclusion?

Answer

A. Long-headed dogs accustomed to hearing Spanish tended to show more brain activity when hearing Spanish than long-headed dogs accustomed to hearing Hungarian showed when hearing Hungarian.

B. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing either Spanish or Hungarian than when hearing the scrambled recording.

C. The pattern of brain activity that long-headed dogs showed when hearing the scrambled recording was different from the pattern of brain activity that short-headed dogs showed when hearing the language they were accustomed to.

D. Compared with shorter-headed dogs, longer-headed dogs showed a greater difference in brain activity when hearing the language they were accustomed to than when hearing the other language.

Correct Answer: B

Rationale

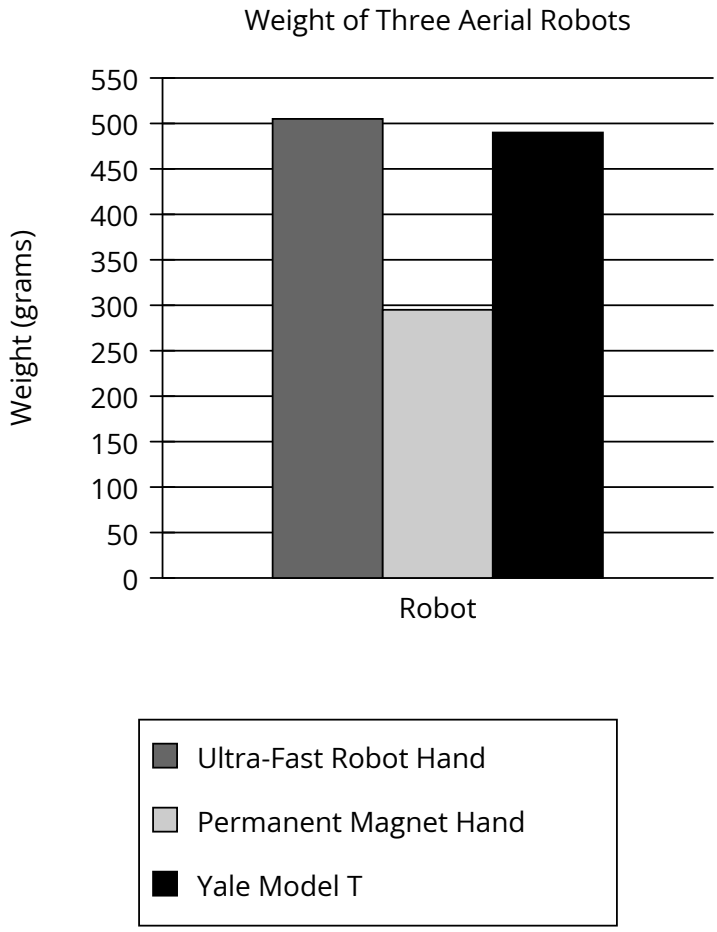
Choice B is the best answer because it presents a finding that, if true, would most directly support the research team's conclusion about anatomical features and speech detection in dogs. The text explains that a team of researchers monitored the brain activity of dogs while the dogs listened to three recordings: one of spoken Spanish, one of spoken Hungarian, and one of scrambled fragments that weren't recognizable as human speech. The text then states that the team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech. The finding that longer-headed dogs exhibited a greater difference in brain activity when listening to the speech recordings (in Spanish or Hungarian) versus the nonspeech (scrambled) recording compared with shorter-headed dogs would establish an association between an anatomical feature (head length) and responses (as measured by brain activity) to speech versus nonspeech. This observed relationship between head length and brain activity patterns during exposure to speech and nonspeech would support the team's conclusion.

Choice A is incorrect because it describes a finding about dogs' responses (as indicated by brain activity) to hearing their respective familiar languages rather than dogs' ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses. Moreover, this finding pertains to only one anatomical type (long-headed dogs), so it wouldn't support the conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. Choice C is incorrect because it describes a finding involving a comparison between long-headed dogs listening to nonspeech and short-headed dogs listening to speech, which wouldn't provide enough information to support the researchers' conclusion that anatomical differences may affect dogs' ability to distinguish speech from nonspeech. To support that conclusion, the finding would need to show how dogs with different anatomical features (for example, long-headed and short-headed dogs) respond to recordings of speech as well as recordings of nonspeech. Choice D is incorrect because it describes a finding about dogs' ability to distinguish between a familiar language and an unfamiliar one, not between speech and nonspeech. While this finding does compare dogs with different anatomical features (longer-headed vs. shorter-headed ones), it focuses on language recognition (Spanish vs. Hungarian) rather than the ability to distinguish speech from nonspeech, which is what the team's conclusion specifically addresses.

Question ID: df37c087

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question



Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot's holding force to the robot's weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues' conclusion?

Answer

A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.

B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.

C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.

D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data in the graph that support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. According to the text, payload capacity is calculated by using a ratio of a robot's holding force to the robot's weight, and higher ratios indicate a greater payload capacity. The Ultra-Fast Robot Hand has a holding force of 56 newtons, four times greater than that of the Yale Model T. Additionally, the graph shows that the Ultra-Fast Robot Hand has a weight of approximately 500 grams, slightly more than the Yale Model T's weight of approximately 480 grams. Therefore, the Ultra-Fast Robot Hand has a higher ratio of holding force to weight than the Yale Model T. Since higher ratios correspond to greater payload capacity, the information from the graph indicating that the Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T combined with the information in the text ultimately supports the conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Choice A is incorrect. Although, according to the graph, it's true that both the Ultra-Fast Robot Hand and the Yale Model T weigh more than 450 grams, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the two robots. Choice B is incorrect. Although, according to the graph, it's true that the Ultra-Fast Robot Hand and the Yale Model T both weigh more than the Permanent Magnet Hand does, this statement doesn't support Meng and colleagues' conclusion that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T. This statement emphasizes a similarity, not a distinction, between the Ultra-Fast Robot Hand and the Yale Model T. Furthermore, the comparison to the Permanent Magnet Hand is irrelevant to the claim about the relative ratios and payload capacities of the Ultra-Fast Robot Hand and the Yale Model T. Choice C is incorrect. Although the text states that the Yale Model T has a lower holding force than the Permanent Magnet Hand, the graph provides no information about holding force. Moreover, information about the Permanent Magnet Hand is irrelevant to the conclusion by Meng and colleagues, which only concerns the Ultra-Fast Robot Hand and the Yale Model T.

Question ID: af9e3240

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____

Which quotation from *Electra* most effectively illustrates the claim?

Answer

A. “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt against this bleeding breast, when dark night fails!”

B. “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”

C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied plaints, while life is in me.”

D. “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of day.”

Correct Answer: C

Rationale

Choice C is the best answer. Electra states that she “knows her own passion,” which shows that she’s aware of the intensity of her grief. But she also claims that the “causes are so dire”—meaning the reasons for her grief are so awful—that she can’t let it go, which shows that she believes her grief is justified.

Choice A is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice B is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice D is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of the intensity of her grief, but it doesn’t suggest that she has a legitimate reason for feeling that way.

Question ID: 2c06139b

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Tadpole Body Mass and Toxin Production after Three Weeks in Ponds

Population density	Average tadpole body mass (milligrams)	Average number of distinct bufadienolide toxins per tadpole	Average amount of bufadienolide per tadpole (nanograms)	Average bufadienolide concentration (nanograms per milligram of tadpole body mass)
High	193.87	22.69	5,815.51	374.22
Medium	254.56	21.65	5,525.72	230.10
Low	258.97	22.08	4,664.99	171.43

Ecologist Veronika Bókony and colleagues investigated within-species competition among common toads (*Bufo bufo*), a species that secretes various unpleasant-tasting toxins called bufadienolides in response to threats. The researchers tested *B. bufo* tadpoles' responses to different levels of competition by creating ponds with different tadpole population densities but a fixed amount of food. Based on analysis of the tadpoles after three weeks, the researchers concluded that increased competition drove bufadienolide production at the expense of growth.

Which choice uses data from the table to most effectively support the researchers' conclusion?

Answer

A. The difference in average tadpole body mass was small between the low and medium population density conditions and substantially larger between the low and high population density conditions.

B. Tadpoles in the low and medium population density conditions had substantially lower average bufadienolide concentrations but had greater average body masses than those in the high population density condition.

C. Tadpoles in the high population density condition displayed a relatively modest increase in the average amount of bufadienolide but roughly double the average bufadienolide concentration compared to those in the low population density condition.

D. Tadpoles produced approximately the same number of different bufadienolide toxins per individual across the population density conditions, but average tadpole body mass decreased as population density increased.

Correct Answer: B

Rationale

Choice B is the best answer. This data shows that the tadpoles in the high-density pond (meaning those with the most competition) didn't grow as big as the other two groups but produced more bufadienolide.

Choice A is incorrect. This doesn't fully support the conclusion. It doesn't include any data about bufadienolide production. Choice C is incorrect. This doesn't fully support the conclusion. It doesn't include any data about growth. Choice D is incorrect. This doesn't fully support the conclusion. It doesn't demonstrate that the tadpoles in the high-density pond produced more bufadienolide overall. The fact that they didn't produce more kinds of bufadienolide isn't relevant to the conclusion.

Question ID: 31ad8024

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Water flowing around an obstruction creates vortices (patterns of swirls) of varying size; by detecting the vortices, fish can determine the size and position of the obstruction. Testing by Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao using models of three head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—showed that for medium-sized vortices, fish with wide heads would be least able to distinguish between vortices and general turbulence in the water. A second research team has therefore hypothesized that in low-visibility conditions, wider-headed fish will be less likely than narrower-headed fish to detect obstructions.

Which finding, if true, would most directly support the second research team's hypothesis?

Answer

A. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the bristlemouth (*Chaetostoma yurubiense*), which has a relatively wide head, bumped into more than half of the obstructions.

B. A study using obstructions that created medium-sized vortices in low-visibility conditions found that some specimens of dusky smooth-hound (*Mustelus canis*), which has a relatively narrow head, bumped into the obstructions more often than other specimens of the same fish did.

C. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (*Chaetostoma yurubiense*) bumped into obstructions more often than the narrower-headed dusky smooth-hound (*Mustelus canis*) did.

D. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the narrower-headed dusky smooth-hound (*Mustelus canis*) bumped into the obstructions just as often as the wider-headed bristlemouth (*Chaetostoma yurubiense*) did.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support the second research team's hypothesis about how a fish's head shape affects the fish's ability to detect obstructions. The text explains that fish detect obstructions using water vortices and that testing by Yanagisuru, Akanyeti, and Liao showed that fish with wide heads struggle to distinguish medium-sized vortices from general turbulence in the water. Based on this finding, a second research team hypothesized that in low-visibility conditions, wider-headed fish would be less effective at detecting obstructions than narrower-headed fish. The finding that wider-headed bristlemouth fish bumped into obstructions more often than narrower-headed dusky smooth-hound fish did under the specified conditions (low visibility with medium-sized vortices) would directly support the second team's hypothesis.

Choice A is incorrect because it would provide only partial support for the second research team's hypothesis. While this finding shows that wide-headed bristlemouth fish bumped into many obstructions, it provides no comparison with narrow-headed fish. The hypothesis specifically predicts a difference in detection ability between fish with wider heads and those with narrower heads, so supporting evidence would need to include data from both head types to allow for this comparison. Choice B is incorrect because it would not support the second research team's hypothesis comparing obstruction detection between fish with wide heads and those with narrow heads. Instead, it examines variation within a single narrow-headed species, which would not address the hypothesis about differences between fish with wider and narrower head shapes. Choice D is incorrect because this finding would contradict rather than support the second research team's hypothesis. The hypothesis predicts that wider-headed fish will be less likely to detect obstructions than narrower-headed fish in low-visibility conditions, which would result in wider-headed fish bumping into obstructions more frequently. A finding showing equal bumping rates between fish with the two head shapes would indicate no relationship between head width and detection ability, directly contradicting the hypothesis.

Question ID: b2e54b50

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Correlations Between Congestion Ratings and Features of the Crowd in Raters' Immediate Vicinity

Crowd feature	Before obstacle	After obstacle	Overall
Density	0.8592	0.7308	0.7447
Velocity	−0.9357	−0.9518	−0.8587

Researcher Xiaolu Jia and colleagues monitored individuals' velocity and the surrounding crowd density as a group of study participants walked through a space and navigated around an obstacle. Participants rated how congested it seemed before the obstacle, after the obstacle, and overall, and the researchers correlated those ratings with velocity and density. (Correlations range from −1 to 1, with greater distance from 0 indicating greater strength). The researchers concluded that the correlations with velocity are stronger than those with density.

Which choice best describes data from the table that support the researchers' conclusion?

Answer

A. The correlation between congestion ratings before the obstacle and density is further from 0 than the correlation between overall congestion rating and velocity is.

B. The correlation between congestion ratings before the obstacle and velocity is further from 0 than the correlation between congestion overall and velocity is.

C. For each of the three ratings, the correlation with velocity is negative while the correlation with density is positive.

D. For each of the three ratings, correlations with velocity are further from 0 than the corresponding correlations with density are.

Correct Answer: D

Rationale

Choice D is the best answer. The text tells us that the farther the correlation is from 0, the "stronger" it is (doesn't matter if it's negative or positive). The table shows that the correlations with velocity are farther from zero than the correlations with density, which supports the conclusion that the correlations with velocity are stronger.

Choice A is incorrect. This choice doesn't support the conclusion. It makes an "apples to oranges" comparison by comparing density and velocity correlations across features instead of comparing them for each feature. Choice B is incorrect. This choice doesn't support the conclusion. It doesn't include the density correlations for comparison. Choice C is incorrect. This choice doesn't support the conclusion. The text tells us that the farther the correlation is from 0, the "stronger" it is: it doesn't matter for "strength" whether it's negative or positive.

Question ID: cca6fae9

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Delta 15-N Values in Seagrass Samples from Four Sites on the Yucatan, 2016–2017

Site	February 2016	October 2016	February 2017	October 2017
Akumal Bay	no data available	3.3	2.0	6.3
Mahahual	0.7	no data available	2.5	3.4
Tulum	6.1	5.9	2.3	5.5
Xahuayxol	0.9	0.3	−0.9	1.4

Because water from natural, uncontaminated sources is less enriched with the stable nitrogen isotope ^{15}N than wastewater from human activities is, the presence of such wastewater in nature can be detected by examining delta 15-N values (a measure of the ratio of ^{15}N to ^{14}N) in plants. Karla A. Camacho-Cruz and colleagues assessed delta 15-N values in the seagrass *Thalassia testudinum* from sites on Mexico's Yucatan peninsula with intermediate tourism development, including Akumal Bay and Tulum, and low tourism development, including Mahahual and Xahuayxol, throughout 2016 and 2017. The data suggest that the intermediate-tourism sites experienced influxes of human wastewater. However, the researchers concluded that this happened intermittently.

Which choice best describes data from the table that support the underlined conclusion?

Answer

A. Although delta 15-N values were generally higher in Akumal Bay and Tulum than in Mahahual and Xahuayxol, the values were lower in Akumal Bay than in Mahahual and Xahuayxol in February 2017.

B. Delta 15-N values reached their lowest level in February 2017 in both Akumal Bay and Tulum, but no data were available for Akumal Bay in February 2016, when the values reached their highest level in Tulum.

C. Although all sites showed considerable variation in delta 15-N values, the values remained relatively constant in Akumal Bay from October 2016 to February 2017 and in Tulum from February 2016 to October 2016.

D. In Akumal Bay and Tulum, delta 15-N values fluctuated considerably across the three measurements made from October 2016 to October 2017.

Correct Answer: D

Rationale

Choice D is the best answer because it describes data from the table that support the underlined conclusion. The text accompanying the table states that delta 15-N values (a measure of the ratio of two nitrogen isotopes) can be used as an indirect means of measuring the ratio of human wastewater to water from uncontaminated sources—the higher the delta 15-N values, the greater the presence of human wastewater. The text goes on to mention an experiment conducted by Camacho-Cruz and colleagues measuring delta 15-N values in seagrass on the Yucatan peninsula at several sites, four of which are presented in the table accompanying the text. Two of the sites noted in the table, Akumal Bay and Tulum, had intermediate tourism development, and two others, Mahahual and Xahuayxol, had low tourism development. It is reasonable to infer that sites with intermediate tourism have a greater human presence than those with low tourism and therefore are subject to greater amounts of wastewater and exhibit higher delta 15-N values in seagrass. However, Camacho-Cruz and colleagues found that the increase in delta 15-N values was not constant. The table supports this conclusion because the two intermediate-tourism sites (Akumal Bay and Tulum) had delta 15-N

values that fluctuated considerably—both had higher delta 15-N values than the two low-tourism sites (Mahahual and Xahuayxol) in October 2016 and October 2017, but in February 2017 the delta 15-N values for the intermediate-tourism sites dropped significantly (in fact, Mahahual, a low-tourism site, had a higher delta 15-N value for that period). The fluctuations in values in Akumal Bay and Tulum support the conclusion that these sites experienced influxes of human wastewater intermittently.

Choice A is incorrect because it inaccurately describes data from the table: in February 2017 the delta 15-N value in Akumal Bay was higher than the delta 15-N value in Xahuayxol, not lower. Choice B is incorrect because the absence of data for delta 15-N levels in Akumal Bay in February 2016 would neither prove nor disprove that the presence of human wastewater at Akumal Bay and Tulum was not constant. Despite the unavailable data, there is still enough information to conclude that delta 15-N levels at both sites fluctuated significantly and that influxes of human wastewater therefore happened intermittently. Choice C is incorrect because a constancy in delta 15-N values would not indicate that the influx of human wastewater was intermittent, but rather the opposite. Moreover, the two periods selected are not indicative of general trends in the whole table (for example, the values fluctuated significantly in Akumal Bay between February 2017 and October 2017, and in Tulum between October 2016 and February 2017).

Question ID: 378c66d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

A member of the Otomi, an Indigenous people in Central Mexico, Octavio Medellín immigrated to the United States as a child, and his sculpture bears the impress of traditions on both sides of the border: US-based modernist sculpture, Mexican modernist painting, Otomi art, and the ancient sculpture of other Mexican Indigenous peoples, including the Maya. In his 1950 masterpiece *History of Mexico*, Medellín fuses these influences into a style so idiosyncratic that it resists efforts to view his work through the lens of nationality or cultural identity. Artists, he insisted, should strive for individual expression, even as they draw inspiration from their heritage and the communities where they live and work.

Which quotation from an art critic most directly challenges the underlined claim in the text?

Answer

A. "Although a number of ancient Indigenous artistic traditions pictured human forms in profile, the forms populating the surface of *A History of Mexico* suggest a specifically Maya influence."

B. "In *A History of Mexico*, the synthesis of ancient and modernist traditions functions as a stylistic parallel to the work's subject matter: a survey of centuries of Mexican history."

C. "Many critics focus on Indigenous influences in *A History of Mexico* and other key works by Medellín to the exclusion of influences from non-Indigenous art."

D. "While *A History of Mexico* features modernist motifs, it relies primarily on angular human forms in profile—a staple of Maya sculpture—and thus invites classification as Indigenous art."

Correct Answer: D

Rationale

Choice D is the best answer. This critic challenges the claim by arguing that *A History of Mexico* is not so idiosyncratic (unique) as to resist classification because its use of Maya-style human profiles actually "invites classification as Indigenous art." Therefore, according to this critic, the work can be viewed, at least partially, through a lens of national or cultural identity.

Choice A is incorrect. While it describes the Maya influence on a particular aspect of *A History of Mexico*, this quotation doesn't push back on Medellín's "idiosyncratic" (unique) style, nor does it try to categorize the entire work into a single cultural tradition. Choice B is incorrect. This choice doesn't directly challenge the underlined claim, but rather supports it. It suggests that the work effectively blends a variety of artistic products to create a single work that can't be defined by any one tradition. Instead, the work recalls many centuries of history and culture all at once. Choice C is incorrect. This choice supports rather than challenges the underlined claim. The quotation argues that other critics focus too much on Indigenous influences on the artwork instead of viewing all of the influences equally—that they should instead be viewing the work as an idiosyncratic whole instead of through one or two narrow cultural lenses.

Question ID: 35ec767c

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Corn-Related Vocabulary in Various Southeastern Languages

Language family	Word (language)	English translation	Proposed origin in vocabulary of the Totozoquean language family
Muskogean	tanchi' (Chickasaw); tanchi (Choctaw); vce (Muscogee, pronounced "uh-chi")	corn	no
Iroquoian	se-lu (Cherokee)	corn	no
Caddoan	-k'as- (Caddo)	dried corn	yes
Chitimacha	k'asma (Chitimacha)	corn	yes

In Caddo, a language from what is now the US Southeast, vocabulary pertaining to corn cultivation resembles equivalent vocabulary in the Totozoquean language family in Mexico. This resemblance is perhaps attributable to cultural contact: such words could have entered Caddo through the intermediary of the neighboring but unrelated Chitimacha language, concurrent with the dissemination of corn itself from Mexico into the Southeast after 700 CE. That the vocabulary pertaining to domestic crops accompanies them as they diffuse into new regions is an established phenomenon globally. Crops may also be decoupled from vocabulary altogether: corn cultivation became ubiquitous among the Southeastern tribes, yet _____

Which choice most effectively uses data from the table to complete the statement?

Answer

A. the origins of vocabulary pertaining to the crop vary across languages in the region, with the words for corn in Cherokee and the Muskogean languages showing no demonstrable relationship to Totozoquean vocabulary.

B. the region is linguistically diverse, being home not only to Chitimacha and Caddo, but also to the Muskogean language family (including Chickasaw, Choctaw, and Muscogee) and to one Iroquoian language (Cherokee).

C. corn-related vocabulary underwent changes when entering other, unrelated languages, as can be seen by the divergence of the Caddo word from the Chitimacha word it originated in.

D. words for corn in the languages of the Muskogean family evolved from a common root, with the Muscogee word having lost certain consonant sounds still present in the Chickasaw and Choctaw words.

Correct Answer: A

Rationale

Choice A is the best answer. The table provides evidence that the words for corn in the Cherokee and Muskogean languages appear unrelated to those in Caddo language, which is described as closely related to the Totozoquean language family. This supports the claim that corn cultivation spread across the Southeast without necessarily spreading Totozoquean vocabulary along with it.

Choice B is incorrect. This choice emphasizes the diversity of Southeastern languages in general, but we're specifically looking for information about words associated with corn. Choice C is incorrect. This choice shows how words for corn can change and evolve, but we're looking for an example that shows how some words for corn can be completely unrelated. Choice D is incorrect. This choice shows words for corn that share a common root, but we're looking for an example that shows how some words for corn can be unrelated.

Question ID: 24c1b7e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Percentage Point Changes in US Federal Outlays Relative to GDP by Congressional Status

Period	Congressional status	Change in total outlays	Change in nondefense outlays	Change in defense outlays
1981–1988	divided	−0.4	−1.3	0.9
1975–1976	divided	2.7	3.0	−0.3
1977–1980	undivided	0.3	0.6	−0.3
1964–1968	undivided	1.9	1.4	0.5
1969–1974	divided	−1.8	2.1	−3.9

Economist Steve H. Hanke has shown that divided US Congresses—which occur when one party holds the majority in the House of Representatives and another holds the majority in the Senate—tend to accompany reductions in total federal outlays (spending) relative to gross domestic product (GDP), which Hanke interprets to reflect decreases in government size. Hanke calculated the percentage point change in total outlays (encompassing nondefense and defense outlays) for consecutive US Congresses. Hanke has pointed to his calculations as evidence that a divided Congress may be a "necessary but not sufficient condition" for a decrease in government size to occur.

Which choice best describes data from the table that support the underlined claim?

Answer

A. The periods of undivided Congresses were associated with increases in nondefense outlays, whereas all the periods of divided Congresses except one were associated with reductions in defense outlays.

B. All the periods of divided Congresses were associated with reductions in total outlays, although two periods were also associated with increases in nondefense outlays.

C. The periods of undivided Congresses were associated with increases in total outlays, whereas all the periods of divided Congresses were associated with reductions in either nondefense outlays or defense outlays.

D. All the periods of divided Congresses except one were associated with reductions in total outlays, whereas the periods of undivided Congresses were associated with increases in total outlays.

Correct Answer: D

Rationale

Choice D is the best answer. The claim is that divided Congresses are necessary but insufficient—that is, we need divide Congresses, but they are not enough—to decrease government size, as measured by total federal outlays. This choice accurately expresses the supporting data from the "change in total outlays" part of the graph. Within the data set, divided Congresses sometimes decreased total outlays, but undivided ones never did.

Choice A is incorrect. The claim is only about government size, as measured by total federal outlays—defense and nondefense outlays aren't relevant. Choice B is incorrect. The claim is only about government size as measured by total federal outlays—nondefense outlays aren't relevant. Choice C is incorrect. The claim is only about government size as measured by total federal outlays—specific information about defense or nondefense outlays isn't relevant.

Question ID: 56f477fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Distribution of Ecosystem Services Affected by Invasive Species by Service Type

Region (Overall)	Provisioning (75%)	Regulating (21%)	Cultural (4%)
West	73%	27%	0%
North	88%	12%	0%
South	79%	14%	7%
East	83%	6%	11%
Central	33%	67%	0%

To assess the impact of invasive species on ecosystems in Africa, Benis N. Egoh and colleagues reviewed government reports from those nations about how invasive species are undermining ecosystem services (aspects of the ecosystem on which residents depend). The services were sorted into three categories: provisioning (material resources from the ecosystem), regulating (natural processes such as cleaning the air or water), and cultural (nonmaterial benefits of ecosystems). Egoh and her team assert that countries in each region reported effects on provisioning services and that provisioning services represent the majority of the reported services.

Which choice best describes data from the table that support Egoh and colleagues' assertion?

Answer

A. Provisioning services represent 73% of the services reported for the West region and 33% of those for the Central region, but they represent 75% of the services reported overall.

B. None of the percentages shown for provisioning services are lower than 33%, and the overall percentage shown for provisioning services is 75%.

C. Provisioning services are shown for each region, while no cultural services are shown for some regions.

D. The greatest percentage shown for provisioning services is 88% for the North region, and the least shown for provisioning services is 33% for the Central region.

Correct Answer: B

Rationale

Choice B is the best answer. The finding that all the regions reported at least some effects on provisioning services supports the first part of the assertion. And the fact that provisioning services comprise 75% of the reported services overall supports the second part of the assertion.

Choice A is incorrect. This choice doesn't fully support the assertion. It doesn't include the finding that all the regions (not just Central and West) reported at least some effects on provisioning services. Choice C is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall. Choice D is incorrect. This choice doesn't fully support the assertion. It doesn't demonstrate that provisioning services represent the majority of the reported services overall.

Question ID: 09775cbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

In a series of experiments, Julio Sevilla and Claudia Townsend showed that manipulating the space between products in store displays can influence consumers' views of those products. Participants in several of the experiments regarded the same products in the same (generic) retail settings as significantly more valuable when the product-to-space ratio was low than when it was high. But in one of the experiments, Sevilla and Townsend arranged the same jewelry with different levels of intervening space at an upscale retailer (Tiffany & Co.) and a relatively inexpensive retailer (Forever 21). The result of this experiment suggests that a store context associated with inexpensive products may moderate the effect Sevilla and Townsend observed in their other experiments.

Which finding from the experiment with Tiffany & Co. and Forever 21, if true, would most directly support the conclusion presented in the text?

Answer

A. At Tiffany & Co., participants judged jewelry spaced far apart to be substantially more valuable than jewelry spaced close together, but at Forever 21, participants judged jewelry spaced far apart to be only slightly more valuable than jewelry spaced close together.

B. At both Tiffany & Co. and Forever 21, participants judged jewelry spaced far apart to be less valuable than jewelry spaced close together, but the difference in perceived value was significantly greater at Tiffany & Co. than at Forever 21.

C. Participants judged jewelry spaced far apart at Tiffany & Co. to be similar in value to jewelry spaced far apart at Forever 21, but participants judged jewelry spaced close together at Tiffany & Co. to be more valuable than jewelry spaced close together at Forever 21.

D. When jewelry was spaced far apart, participants judged the jewelry at Tiffany & Co. to be more valuable than the jewelry at Forever 21, but when jewelry was spaced close together, participants judged the jewelry at Tiffany & Co. to be less valuable than the jewelry at Forever 21.

Correct Answer: A

Rationale

Choice A is the best answer because it presents a finding that, if true, would support the text's conclusion about the results of the experiment with Tiffany & Co. and Forever 21. According to the text, Sevilla and Townsend found in several experiments that when products in a generic retail setting are displayed with a low product-to-space ratio (that is, with lots of space between them), consumers view those products as significantly more valuable than when the same products are displayed with a higher product-to-space ratio (less space between them). The text then states that the results of an experiment specifically using the contexts of an inexpensive store (Forever 21) and an upscale one (Tiffany & Co.) suggest that an inexpensive store context may moderate, or lessen, that effect. If Sevilla and Townsend found that participants in the experiment with Tiffany & Co. and Forever 21 judged the same jewelry items as substantially more valuable when there was lots of space between them than when there was little space between them in the upscale store (Tiffany & Co.) but judged the same jewelry items as only slightly more valuable when there was lots of space between them than when there was little space between them in the inexpensive store (Forever 21), that finding would demonstrate that increased space between products was associated with less of an increase in those products' perceived value at the inexpensive store than at the upscale store. Thus, the finding would support the text's conclusion that a store context associated with inexpensive goods moderates the effect Sevilla and Townsend observed in their earlier experiments.

Choice B is incorrect because if Sevilla and Townsend found that at both upscale and inexpensive stores, participants judged jewelry spaced far apart to be less valuable than jewelry spaced close together, that would show the opposite of the effect the researchers observed in their earlier experiments, not show that an inexpensive store context merely moderates, or lessens, that effect. Choice C is incorrect because this finding wouldn't show that the effect Sevilla and Townsend observed in their initial experiments (that products spaced far apart were perceived as more valuable than the same products spaced close together) is present but lessened in inexpensive retail contexts, as the text suggests. The conclusion in the text rests on determining the difference in jewelry items' perceived value between two spacing conditions within each store and then comparing the difference for the inexpensive store (Forever 21) to the difference for the upscale store (Tiffany & Co.); a finding that

compares perceptions of the jewelry items' value between the two stores but for only one type of spacing condition at a time wouldn't provide information about the degree of difference between spacing conditions within each type of store context. Choice D is incorrect because this finding wouldn't show that the effect Sevilla and Townsend observed in their initial experiments (that products spaced far apart were perceived as more valuable than the same products spaced close together) is present but lessened in inexpensive retail contexts, as the text suggests. The conclusion in the text rests on determining the difference in jewelry items' perceived value between two spacing conditions within each store and then comparing the difference for the inexpensive store (Forever 21) to the difference for the upscale store (Tiffany & Co.); a finding that compares perceptions of the relative value of jewelry items between the two stores and for only one type of spacing condition at a time wouldn't provide information about the degree of difference between spacing conditions within each type of store context.

Question ID: 8391a002

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Black beans (*Phaseolus vulgaris*) are a nutritionally dense food, but they are difficult to digest in part because of their high levels of soluble fiber and compounds like raffinose. They also contain antinutrients like tannins and trypsin inhibitors, which interfere with the body's ability to extract nutrients from foods. In a research article, Marisela Granito and Glenda Álvarez from Simón Bolívar University in Venezuela claim that inducing fermentation of black beans using lactic acid bacteria improves the digestibility of the beans and makes them more nutritious.

Which finding from Granito and Álvarez's research, if true, would most directly support their claim?

Answer

A. When cooked, fermented beans contained significantly more trypsin inhibitors and tannins but significantly less soluble fiber and raffinose than nonfermented beans.

B. Fermented beans contained significantly less soluble fiber and raffinose than nonfermented beans, and when cooked, the fermented beans also displayed a significant reduction in trypsin inhibitors and tannins.

C. When the fermented beans were analyzed, they were found to contain two microorganisms, *Lactobacillus casei* and *Lactobacillus plantarum*, that are theorized to increase the amount of nitrogen absorbed by the gut after eating beans.

D. Both fermented and nonfermented black beans contained significantly fewer trypsin inhibitors and tannins after being cooked at high pressure.

Correct Answer: B

Rationale

Choice B is the best answer because it presents a finding that would best support Granito and Álvarez's claim that fermenting black beans makes them easier to digest and more nutritious. The text indicates that high levels of soluble fiber and raffinose in black beans make the beans hard to digest and that tannins and trypsin inhibitors make it harder for the body to extract nutrients from the beans. If it were found that fermenting the beans significantly reduces their levels of soluble fiber, raffinose, trypsin inhibitors, and tannins when cooked, this would directly support the claim that fermentation improves the digestibility of the beans and makes them more nutritious.

Choice A is incorrect because the text indicates that trypsin inhibitors and tannins interfere with the body's ability to extract nutrients from black beans; if fermentation and cooking were found to increase these antinutrients, fermented beans would likely be less nutritious than unfermented ones, not more nutritious (as Granito and Álvarez claim). Choice C is incorrect because the text doesn't address the idea that greater nitrogen absorption in the gut has an effect on a food's digestibility or level of nutrition, so the discovery of the presence of microorganisms that may increase nitrogen absorption wouldn't provide relevant support for the claim that fermentation makes black beans easier to digest and more nutritious. Choice D is incorrect because Granito and Álvarez's claim focuses on the effect of fermenting black beans, but the finding that nonfermented black beans also have fewer trypsin inhibitors and tannins when cooked at high pressure would suggest that the role of the cooking method could be significant when it comes to nutrition; further, the finding wouldn't address the beans' digestibility.

Question ID: 124fdcd7

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Many archaeologists will tell you that categorizing excavated fragments of pottery by style, period, and what objects they belong to relies not only on standard criteria, but also on instinct developed over years of practice. In a recent study, however, researchers trained a deep-learning computer model on thousands of images of pottery fragments and found that it could categorize them as accurately as a team of expert archaeologists. Some archaeologists have expressed concern that they might be replaced by such computer models, but the researchers claim that outcome is highly unlikely.

Which finding, if true, would most directly support the researchers' claim?

Answer

A. In the researchers' study, the model was able to categorize the pottery fragments much more quickly than the archaeologists could.

B. In the researchers' study, neither the model nor the archeologists were able to accurately categorize all the pottery fragments that were presented.

C. A survey of archaeologists showed that categorizing pottery fragments limits the amount of time they can dedicate to other important tasks that only human experts can do.

D. A survey of archaeologists showed that few of them received dedicated training in how to properly categorize pottery fragments.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text explains that although archaeologists hold that categorizing pottery fragments relies on both objective criteria and instinct developed through direct experience, researchers have found that a computer model can categorize the fragments with the same degree of accuracy as the humans can—a finding that has caused some archaeologists to worry that their own work won't be needed any longer. If survey results indicate that categorizing pottery fragments limits the amount of time archaeologists can dedicate to other important tasks that only human experts can do, that would mean that computer models aren't able to do all of the important things archaeologists do, thus supporting the researchers' claim that computer models are unlikely to replace human archaeologists.

Choice A is incorrect because if it were true that the computer model could categorize the pottery fragments much more quickly than the archaeologists could, that would weaken the researchers' claim that archaeologists are unlikely to be replaced by certain computer models, since it would demonstrate that the models could conduct the archaeologists' work not only with equal accuracy but also at a faster pace. Choice B is incorrect because the inability of both the computer model and the archaeologists to accurately categorize all of the pottery fragments presented wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text indicates that some archaeologists are worried because the computer model's accuracy is equal to their own, and that could be the case whether both were perfectly accurate or were unable to achieve complete accuracy. Choice D is incorrect because survey results showing that few archaeologists received special training in properly categorizing pottery fragments wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The amount of special training in categorizing pottery fragments that archaeologists have received has no direct bearing on whether computer models' success at categorizing fragments will lead to the models replacing the archaeologists.